SESSION 6: NLP WITH MACHINE LEARNING AND DEEP LEARNING

Theory and Laboratory (2T + 2SL) | List of 10 Recommended Laboratory Programs

The following laboratory programs cover the Session 6 topics, progressing from traditional Machine Learning methods to Deep Learning applications in Natural Language Processing (NLP).

List of 10 Laboratory Programs

Sl. No.	Laboratory Program	Concepts / Algorithms Covered
1	**Implementation of Text Classification using Machine Learning**	Text preprocessing, Bag of Words, TF-IDF and feature extraction
2	**Implementation of Naive Bayes Classifier for Spam Email Detection**	Multinomial Naive Bayes, probability and text classification
3	**Sentiment Analysis using Logistic Regression and Support Vector Machine (SVM)**	Logistic Regression, SVM, TF-IDF, positive and negative sentiment classification
4	**Comparison of Machine Learning Algorithms for NLP Tasks**	Naive Bayes, Decision Tree, Random Forest and Logistic Regression; accuracy, precision, recall and F1-score
5	**Implementation of Word Embeddings for NLP using Word2Vec**	One-hot encoding, word vectors, CBOW, Skip-gram and semantic similarity
6	**Implementation of Neural Network for Text Classification**	Artificial Neural Network (ANN), input/hidden/output layers, activation functions and backpropagation
7	**Implementation of Recurrent Neural Network (RNN) for Text Sequence Processing**	RNN architecture, sequential data processing, hidden states and vanishing-gradient problem
8	**Implementation of Long Short-Term Memory (LSTM) for Sentiment Analysis**	LSTM architecture, input gate, forget gate, output gate, cell state and hidden state
9	**Next Word Prediction using LSTM Networks**	Tokenization, sequence generation, padding, embedding layer, LSTM and text generation
10	**Comparative Analysis of Machine Learning and Deep Learning Models for NLP**	Naive Bayes, SVM, ANN, RNN and LSTM; performance comparison and visualization

Recommended Datasets and Python Libraries

Program	Suggested Dataset	Python Libraries
1	SMS Spam Collection	NLTK, Scikit-learn
2	SMS Spam Collection	Pandas, Scikit-learn
3	IMDb Movie Reviews	NLTK, Scikit-learn
4	20 Newsgroups Dataset	Scikit-learn, Matplotlib
5	Text corpus / Wikipedia sentences	Gensim, NLTK
6	IMDb Movie Reviews	TensorFlow, Keras
7	Shakespeare text dataset	TensorFlow, Keras
8	IMDb Movie Reviews	TensorFlow, Keras
9	Shakespeare text dataset	TensorFlow, Keras
10	IMDb Movie Reviews	Scikit-learn, TensorFlow, Matplotlib

Mapping of Laboratory Programs to Session 6 Theory

Theory Topic	Relevant Laboratory Coverage
Machine Learning and Deep Learning for NLP	Programs 1–6: text classification, Naive Bayes, Logistic Regression, SVM, Word2Vec and ANN.
ML vs. DL and respective algorithms	Programs 4 and 10: compare traditional ML and DL methods using evaluation metrics.
Linear Algebra Review	Programs 1, 5 and 6: vectors, matrices, matrix multiplication, weight matrices and vector representations of words.
Long Short-Term Memory (LSTM)	Programs 8 and 9: LSTM architecture, memory cells, gates, sentiment analysis and next-word prediction.

Recommended Software Setup

• Python 3.x • Jupyter Notebook or Google Colab • NLTK and Scikit-learn • TensorFlow / Keras • Gensim • Pandas and NumPy • Matplotlib and Seaborn

Suggested Implementation Sequence

• **Programs 1–4:** Traditional Machine Learning for NLP.

• **Programs 5–6:** Word Embeddings and Neural Networks.

• **Programs 7–9:** RNN and LSTM applications.

• **Program 10:** Comparative analysis of Machine Learning and Deep Learning.

Teaching emphasis: Give special attention to Programs 8 and 9 because LSTM is explicitly included in the syllabus. Before implementation, ensure students understand the forget gate, input gate, output gate and cell state.

Suggested lab-manual format: Each experiment can be expanded with an aim, learning objectives, brief theory, requirements, step-by-step execution, Python source code, sample output, result and viva questions.